

THUSO NDOU

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PROFESSIONAL SUMMARY

BSc Information Technology graduate from North-West University, currently pursuing a BSc Honours in Data Science at Eduvos. Founder and Software Developer at Knolink with hands-on experience developing full-stack applications, AI systems, data-driven solutions, APIs and cloud applications. Experienced across software development, data analysis, artificial intelligence and cloud computing, with practical experience using Python, Java, TypeScript, React, Node.js, SQL, MongoDB and Microsoft Azure.

SKILLS

- AI & Machine Intelligence: AI Application Development, Microsoft Foundry, Multi-Agent Systems, AI Inference, Reinforcement Learning, OpenAI APIs, Vector Databases, Embedding Pipelines
- Cloud & Infrastructure: Microsoft Azure, Azure App Service, Cloud Computing, Cloud-Native Application Development, Docker, Application Deployment, Linux
- Software Engineering: Python, TypeScript, JavaScript, Java, C#, React, Node.js, Express.js, REST API Development, Software Architecture, Full-Stack Development
- Data & Databases: Data Analysis, Data Modelling, SQL, MongoDB, Database Design & Management
- Security & Systems: Authentication & Access Control, Secure Coding, WebSocket, Server-Sent Events, System Analysis & Design, API Development & Integration
- Tools: Git, GitHub, Tailwind CSS

EDUCATION

Bachelor of Science in Information Technology	Jan 2022 – Dec 2025
North-West University	
Completed	
Bachelor of Science Honours in Data Science	Jul 2026 – Present
Eduvos	
In Progress	

WORK EXPERIENCE

Founder & Software Developer – Knolink	Jul 2025 – Present
• Identified a gap in the academic research workflow and developed a product strategy around AI and cloud-based solutions. • Designed and developed the platform end-to-end, including frontend, backend, database architecture and AI integrations. • Designed and implemented secure authentication, data validation and cloud deployment using Microsoft Azure. • Manage the product lifecycle from ideation and development through deployment, user onboarding and continuous improvement. • Engage directly with students and researchers to gather feedback and translate user requirements into product improvements. • Research and implement emerging AI and cloud technologies to improve platform capabilities and performance.	

- Implemented secure coding practices, including password policies and access controls.
- Developed and maintained web and mobile applications for multiple clients.
- Built and integrated REST APIs, frontend interfaces and database solutions using Python, Java, C# and JavaScript.
- Collaborated with clients to gather requirements and deliver functional, user-focused solutions.
- Debugged and resolved software issues during development and testing.
- Assisted with system configuration and application deployment in development environments.
- Automated internal processes to improve development and operational efficiency.

PROJECTS

Roots Online Butchery Platform - Personal Project 2024

A complete end-to-end e-commerce platform for a butchery business, featuring Stripe payment processing, JWT token-based authentication, real-time inventory management, MongoDB aggregation pipelines and a comprehensive admin dashboard.

Role: Full-Stack Developer.

End-to-end platform development, including the React frontend, Node.js/Express REST API, Stripe Checkout webhook integration, role-based access control, MongoDB data modelling.

Technologies: React, Node.js, Express, MongoDB, Stripe API, JWT, Tailwind CSS, Git

Client Web & Mobile Application Development - BlueGrid Digitals 2024

Developed and maintained web and mobile applications for various clients, including frontend interfaces, backend APIs and database solutions.

Role: Developer.

Built and integrated REST APIs, developed frontend interfaces, collaborated with clients to understand requirements and delivered functional, user-friendly solutions.

Technologies: Python, Java, C#, JavaScript, Node.js, Git, Linux, Windows

Knolink Machine Intelligence Suite - Personal Project 2025

Unified AI infrastructure for multi-agent inference, coordinating inference runtimes, vector retrieval, and autonomous reinforcement learning across specialized compute nodes. Focused on reliable, scalable and low-latency processing.

Role: Founder and Lead Engineer.

Designed and developed the system end-to-end, including the distributed orchestration layer, vector retrieval pipeline, embedding workflows and AI integrations.

Technologies: Azure App Service, Docker, TypeScript, Pinecone, OpenAI API, Firebase, Tailwind CSS, Node.js, Git, Microsoft Foundry

Apolemia – Knolink 2025

A high-throughput inference execution engine designed for real-time model serving. Features dynamic KV-cache reuse, activation sparsity routing, enclave memory security, and low-latency streaming token distribution.

Role: Lead Developer and Systems Architect.

Designed and implemented the inference runtime, including the dynamic KV-cache reuse pipeline, routing logic, resilient WebSocket and SSE token streaming dispatch

Technologies: TypeScript, Python, React, Node.js, WebSocket, Server-Sent Events (SSE), Tailwind CSS, Git

Entropy – Knolink

2026

A high-performance developer IDE and AST compiler environment designed for deep code intelligence. Built with modular Rust crates and a reactive TypeScript interface for fast syntax processing and deterministic code editing.

Role: Lead Developer and Compiler Engineer.

Designed the modular Rust crate architecture, built the AST parser and static analysis engine and developed the deterministic multi-pane editing canvas.

Technologies: TypeScript, React, WebAssembly, Tailwind CSS, Git

Praxis – Knolink

2026

An experimental reinforcement learning environment studying how strategic planning emerges from autonomous feedback, deep Monte Carlo Tree Search (MCTS) and value network convergence.

Role: Lead Researcher and Developer.

Developed the reinforcement-learning environment, implemented Monte Carlo Tree Search (MCTS) and self-play mechanisms, and evaluated model behaviour and learning progression.

Technologies: Python, Reinforcement Learning, MCTS, Git

Rudmak Chilli Sauce E-Commerce Platform – Rudmak

2026

Developed a responsive e-commerce and product-ordering platform for Rudmak Chilli Sauce, featuring product catalogues, configurable product options, interactive cart functionality and direct WhatsApp checkout.

Role: Web Developer.

Designed and developed the responsive frontend, product catalogue, interactive shopping cart, order builder and WhatsApp checkout workflow. Implemented product and customer information handling, dynamic order summaries, mobile navigation, FAQ components, product pairing sections and SEO-optimised page structure.

Technologies: HTML, CSS, JavaScript, WhatsApp Integration, Schema.org, Git

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Sep 2022 — Cisco Introduction to Cybersecurity (Cisco)

Dec 2022 — MTN Business App Academy Intermediate-POE (Certificate of completion) (MTN)

Mar 2023 — IBM Data Science Foundations- Level1 (IBM)

Apr 2023 — IBM Data Science Tools (IBM)

2023 — Golden Key International Honour Society Member (Golden Key International Honour Society)

Jul 2024 — Ayoba Business Chat API Hackathon (Ayoba)

Dec 2024 — Chongqing Polytechnic University of Electronic Technology Basic Training on Network Security Management (Chongqing Polytechnic University of Electronic Technology)

PORTFOLIO & LINKS

Knolink — <https://knolink.co.za>

GitHub — <https://github.com/Thuso-Ndou>

Personal Website — <https://virtual-cv-roan.vercel.app/>

Praxis — <https://praxis.knolink.co.za>

Apolemia — <https://apolemia.knolink.co.za>

Entropy — <https://entropy.knolink.co.za>

Rudmak — <https://rudmak.co.za>

REFERENCES

Tibo — Former Lecturer
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Successfull — IT Manager at BlueGrid Digital
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